ARJUNA JEE 2027

Lecture -08 | Mathematics

PERMUTATIONS AND COMBINATIONS

By- Sachin Jakhar Sir



Topics *to be covered*

A Selection out of things which are not all different

B Divisor Problems

C Question Practice

D



H.W. Discussion

Acidic Question



[Ans. : 210]

Find the number of 7 digit integers whose sum of digits is 59.

$_ + _ + _ + _ + _ + _ + _ = 59$

C-I : $9,9,9,9,9,\overset{14}{\overbrace{9,5}} \rightarrow \frac{7!}{6!} = 7$

C-II : $9,9,9,9,9,\overset{14}{\overbrace{8,6}} \rightarrow \frac{7!}{5!} = 42$

C-III : $9,9,9,9,9,\overset{14}{\overbrace{7,7}} \rightarrow \frac{7!}{5!2!} = 21.$

C-IV : $\overbrace{9,9,9,9}^{36},\overbrace{8,8,7}^{23} \rightarrow \frac{7!}{4!2!} = 105.$

C-V : $\overbrace{9,9,9}^{27}\ \overbrace{8,8,8,8}^{32} \rightarrow \frac{7!}{3!4!} = 35.$

Question

[Ans. : D]

The number of 5-digit ATM passwords having at least one of their digits repeated is

A 90,000

B 10,000

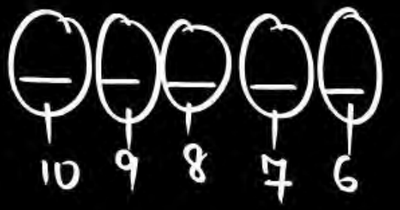
C 30,240

D 69,760

01234

Total = _ _ _ _ _ → 10^5 − (No digit repeated)

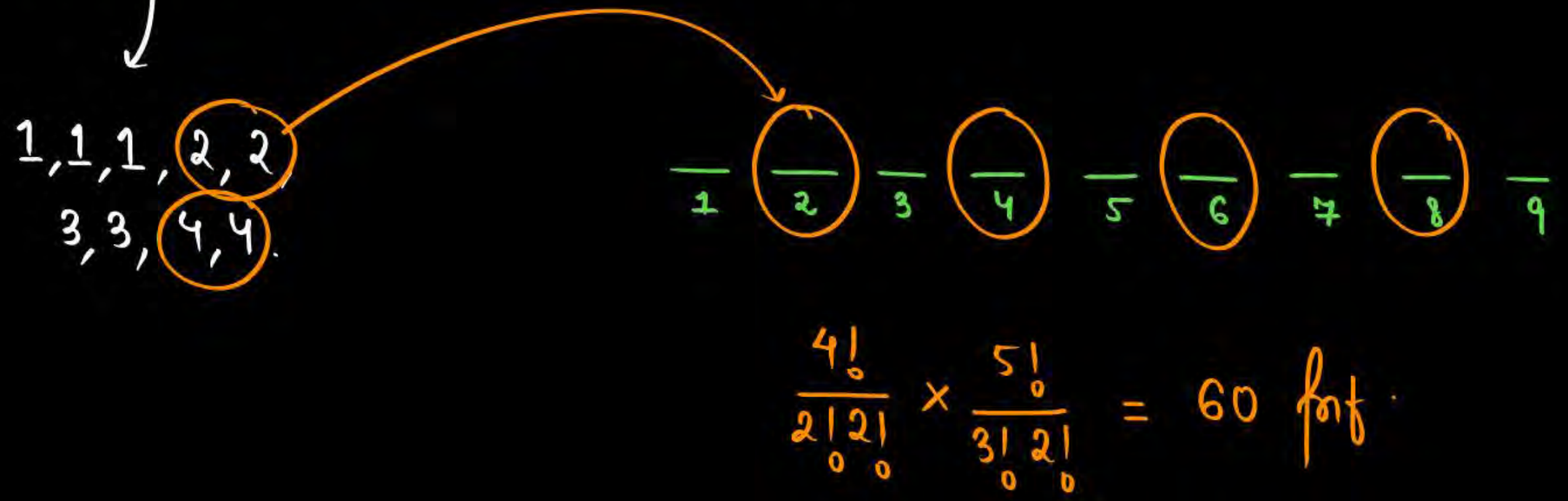
10 10 10 10 10



fnf = $10^5 - (10 \times 9 \times 8 \times 7 \times 6)$

Question [JEE Mains-2023]

The number of 9 digit numbers that can be formed using all the digits of the number 123412341 so that the even digits occupy only even places, is _____.



Reverse Question

If all the letters of the word **PUBLIC** be permuted and all the words so formed (with or without meaning) be listed as in a dictionary, then word written at 437th position??

~~B~~, C, I, ~~L~~, ~~P~~, ~~U~~.

B _ _ _ _ _ → 5! = 120.

C _ _ _ _ _ → 5! = 120 — 240.

I _ _ _ _ _ → 5! = 120 — 360

L B _ _ _ → 4! = 24 — 384

L C _ _ _ → 4! = 24 — 408.

L I _ _ _ → 24 — 432

L P B C _ _ → 2

L P B I _ _ → 2 — 436.

L P B U C I → 1 — 437.

Question [JEE Mains-2016]

If all the words (with or without meaning) having five letters, formed using the letters of the word SMALL and arranged as in a dictionary; then the position of the word SMALL is

[Ans. : (C)]

A 59th

B 52nd

C 58th

D 46th

Question [JEE Mains-2023]

If the letters of the word MATHS are permuted and all possible words so formed are arranged as in a dictionary with serial numbers, then the serial number of the word THAMS is

[Ans. : (A)]

A 103

B 104

C 102

D 101

Question

If the letters of the word SACHIN are arranged in all possible ways and these words are written out as in dictionary, then the word SACHIN appears at serial number

[Ans. : (D)]

A 602

B 603

C 600

D 601

A, C, H, I, N, S

QIBY

If all the six digit numbers $x_1\,x_2\,x_3\,x_4\,x_5\,x_6$ with $0 < x_1 < x_2 < x_3 < x_4 < x_5 < x_6$ are arranged in the increasing order, then the sum of the digits in the 72th number is_____

$1_____ \rightarrow {}^{8}C_5 \times 1 = 56$. $\{1,2,3,4,5,6,7,8,9\}$

$2_____ \rightarrow {}^{7}C_5 \times 1 = 21$

$23____ \rightarrow {}^{6}C_4 \times 1 = 15$

$2456\,7\,8 \rightarrow 1$ — 72th

Sum $= (32)$ Ans.

ex: 134789 ✓

125689 ✓

123456 ✓

123564 ✗

SELECTION OUT OF THINGS WHICH ARE NOT ALL DIFFERENT

Example

Number of ways of selecting three letters out A, B, C, D, E.

$^5C_3$ = 10 ways.

SELECTION OUT OF THINGS WHICH ARE NOT ALL DIFFERENT

Example

not all different.

Number of ways of selecting three letters out AAA, BB, C, D

3 letters

I (all diff) (ABC, BCD, ...) → A, B, C, D. → $^4C_3$

II (2 alike + 1 diff) (AAB, BBC, ACA ....) → (AA), (BB) → $^2C_1 \times {}^3C_1 = 6$

2 alike 1 diff

ABC
ACD
ABD
BCD

AAB
AAC
AAD
BBA
BBC
BBD.

III (3 alike) (AAA) → 1

AAA.

Total ways = 4 + 6 + 1.

= 11

INDEPENDENCE → select any 5 letters. $= (6 + 30 + 20 + 5 + 12 + 4)$

EEEE NNN DD I P C

C-I : All different (ENDIP, NDIPC, ...) → ${}^{6}C_{5} = 6$

C-II : 2-alike + 3 different (EENDI, NNEPC) → ${}^{3}C_{1} \times {}^{5}C_{3} = 30$

C-III : 3-alike + 2 diff (EEEDC, NNNEI....) → ${}^{2}C_{1} \times {}^{5}C_{2} = 20$

C-IV : 4-alike + 1 diff (EEEED...) → $1 \times {}^{5}C_{1} = 5$

INDEPENDENCE → select any 5 letters.

EEEE NNN DD I P C

C-V : 2-alike + 2-other alike + 1 diff. → ${}^{3}C_{2} \times {}^{4}C_{1} = 12$

(EENNI, NNDDC...)

C-VI : 3-alike + 2-other alike. → ${}^{2}C_{1} \times {}^{2}C_{1} = 4$

(EEENN, NNNDD...)

{EEENN, EEEDD, NNNEE, NNNDD.}

form a 5 letter word using letters of the word INDEPENDENCE

EEEE NNN DD I P C

$\Rightarrow$ Req = p + q + r + l + m + n

Select^{n}. $\times$ arrang.

C-I : All different (ENDIP, NDIPC, ...)

$\Rightarrow {}^{6}C_{5} \times 5!_{0} = p$

C-II : 2-alike + 3 different (EENDI, NNEPC)

$\Rightarrow {}^{3}C_{1} \times {}^{5}C_{3} \times \frac{5!_{0}}{2!_{0}} = q$

C-III : 3-alike + 2 diff (EEEDC, NNNEI....)

$\Rightarrow {}^{2}C_{1} \times {}^{5}C_{2} \times \frac{5!_{0}}{3!_{0}} = r$

C-IV : 4-alike + 1 diff. (EEEED...)

$\Rightarrow 1 \times {}^{5}C_{1} \times \frac{5!_{0}}{4!_{0}} = l$

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C-V :- 2-alike + 2-other alike + 1 diff.

(EENNI, NNDDC...)

$\rightarrow {}^{3}C_{2} \times {}^{4}C_{1} \times \frac{5!}{2!2!} = m$

C-VI :- 3-alike + 2-other alike.

(EEENN, NNNDD...)

$\rightarrow {}^{2}C_{1} \times {}^{2}C_{1} \times \frac{5!}{3!2!} = n$

{EEENN, EEEDD, NNNEE, NNNDD.}

Question [JEE Mains-2020]

The number of 4 letter words (with or without meaning) that can be formed from the eleven letters of the word EXAMINATION is _____.



Question [JEE Mains-2026]

[Ans. : (D)]

The number of 4-letter words, with or without meaning, each consisting of two vowels and two consonants that can be formed from the letters of the word INCONSEQUENTIAL, without repeating any letter, is



A 2670

B 2840

C 2920

D 3600



ALL POSSIBLE SELECTIONS

'n'- Different Things

number of ways of selecting Zero thing = ${}^nC_0$

number of ways of selecting One thing = ${}^nC_1$

number of ways of selecting Two things = ${}^nC_2$

.

.

.

number of ways of selecting All things = ${}^nC_n$

add

$${}^nC_0 + {}^nC_1 + {}^nC_2 + \ldots\ldots + {}^nC_n = 2^n$$

'n'- Identical Things

number of ways of selecting Zero thing = 1

number of ways of selecting One thing = 1

number of ways of selecting Two things = 1

.

.

.

number of ways of selecting All things = 1

add $\Rightarrow$ $(n+1)$

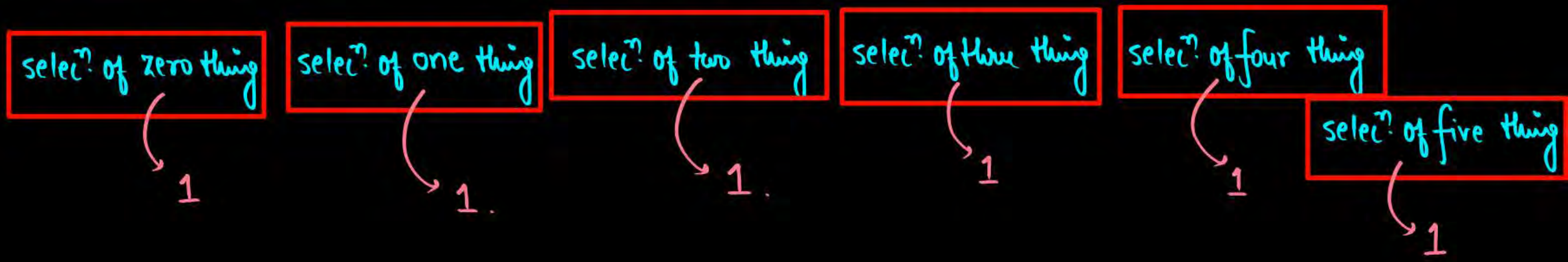
$S = \{1,2,3,4,5\}$

Subset $\longrightarrow 2^5 = 32$.

selecⁿ of zero thing	selecⁿ of one thing	selecⁿ of two thing	selecⁿ of three thing	selecⁿ of four thing	selecⁿ of five thing
${}^5C_0$	${}^5C_1 = 5$	${}^5C_2 = 10$	${}^5C_3 = 10$	${}^5C_4 = 5$	${}^5C_5 = 1$
1					

$${}^5C_0 + {}^5C_1 + {}^5C_2 + {}^5C_3 + {}^5C_4 + {}^5C_5 = 2^5.$$

AAAAA


selecⁿ of zero thing
1
selecⁿ of one thing
1
selecⁿ of two thing
1
selecⁿ of three thing
1
selecⁿ of four thing
1
selecⁿ of five thing
1

Bumper Question

From a library containing 4 different physics book, 5 identical chemistry books & 6 identical maths books. Find the number of ways in which we can select

(i) Atleast 1 book $= 2^4 \cdot 6 \cdot 7 - (1)$ (zero book selected)

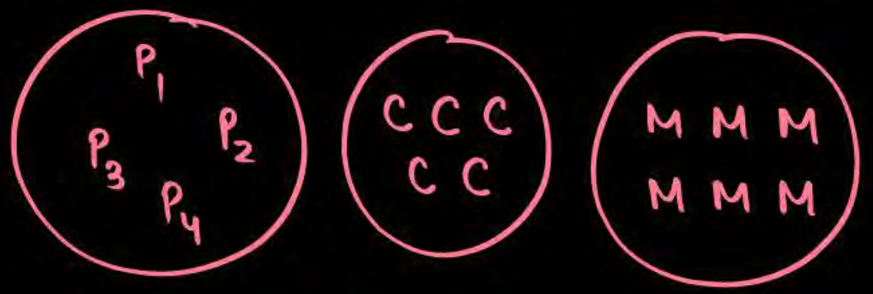
(ii) Atleast 1 book of each subject $= (2^4 - 1)\, 5 \cdot 6$

(iii) Atleast 2 books $= 2^4 \cdot 6 \cdot 7 - \left(\underset{\text{zero book}}{1} + \underset{\text{1 book}}{\left({}^4C_1 + 1 + 1\right)}\right)$

(iv) Atleast 2 books of each subject

(v) Exactly 2 books

(vi) Exactly 2 books of each subject.



All possible selec^n. $= 2^4 \times (5+1) \times (6+1)$

$\left({}^4C_2 + {}^4C_3 + {}^4C_4\right) \times (1+1+1+1) \times (1+1+1+1+1)$

$\hookrightarrow \left(2^4 - {}^4C_0 - {}^4C_1\right) \times (4) \times (5)$

Bumper Question

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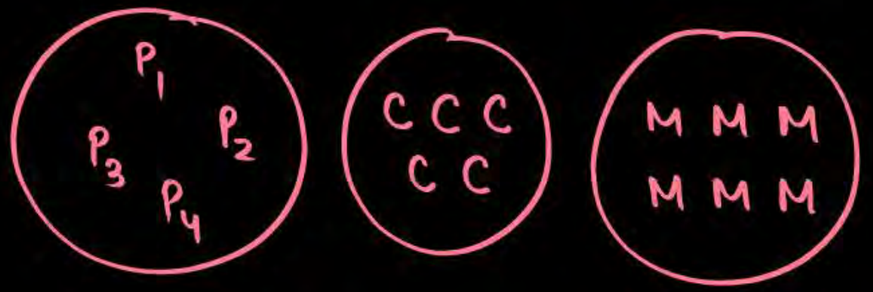
(ii) Atleast 1 book of each subject $= (2^4 - 1) \cdot 5 \cdot 6$

(iii) Atleast 2 books $= 2^4 \cdot 6 \cdot 7 - (\underset{\text{zero book}}{1} + (\underset{\text{1 book}}{{}^4C_1} + 1 + 1))$

(iv) Atleast 2 books of each subject

(v) Exactly 2 books

(vi) Exactly 2 books of each subject.

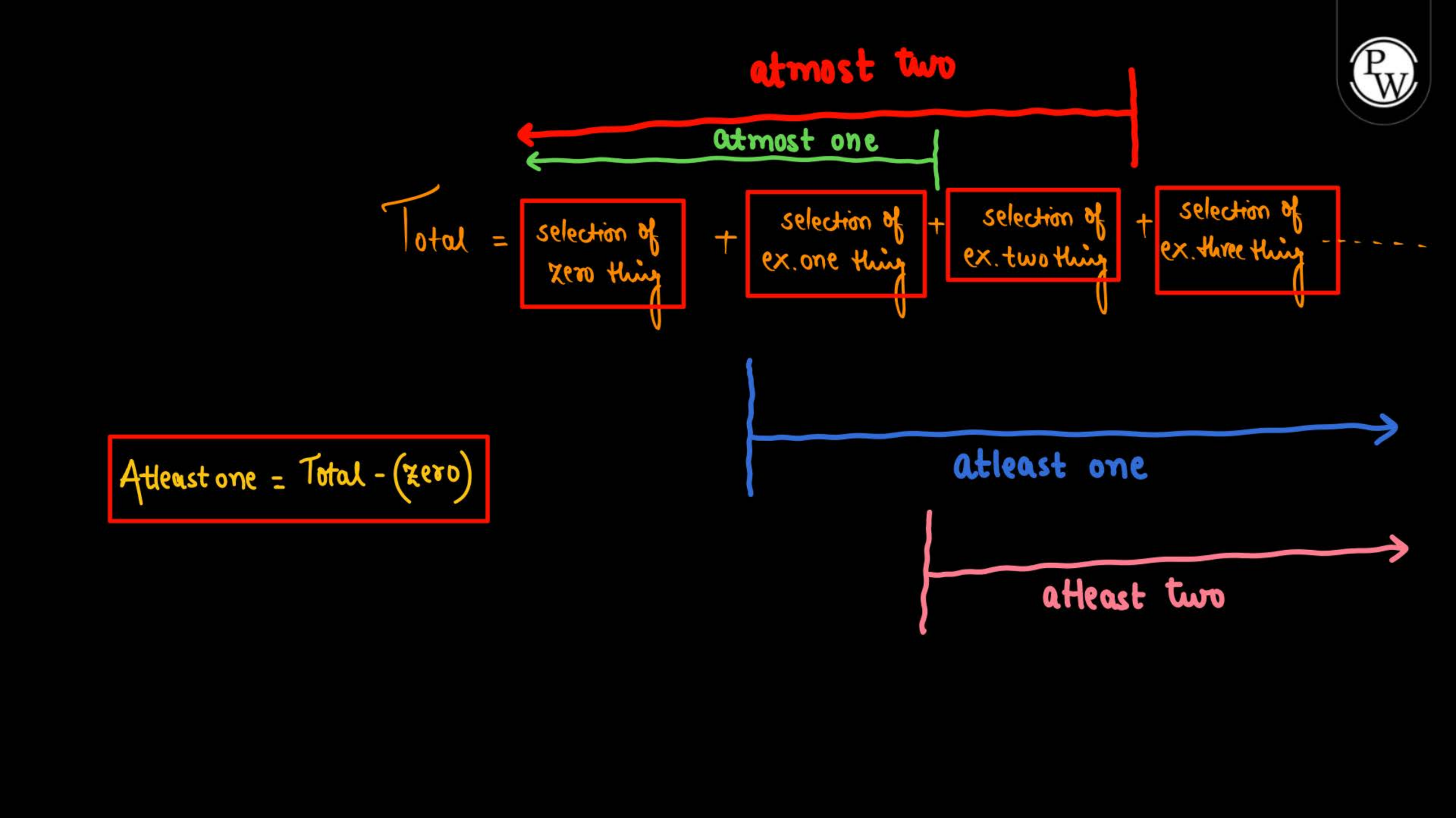


OR $\equiv$ (Atleast three books − Atleast two books)

(iv) PP + CC + MM + PC + PM + CM

${}^4C_2 + 1 + 1 + {}^4C_1 \times 1 + {}^4C_1 \times 1 + 1 \times 1 = 17$ ways.

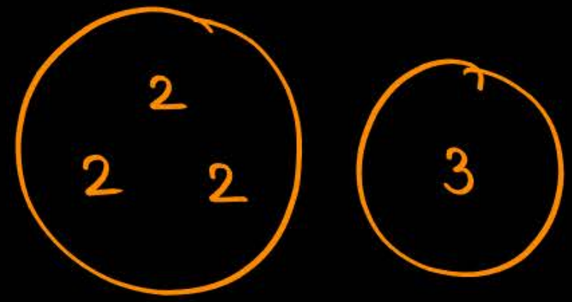
(vi) ${}^4C_2 \times 1 \times 1 = 6$


atmost two
atmost one
Total = selection of zero thing + selection of ex. one thing + selection of ex. two thing + selection of ex. three thing
Atleast one = Total - (zero)
atleast one
atleast two

$N = 24 = 2^3 \cdot 3^1$



→ Total = 8 div.



all possible selec^n. $= (3+1)\cdot(1+1)$

$= 4 \times 2$

$= 8$

$N = 2^7 \cdot 3^6 \cdot 5^4$

↳ No. of div. $= (7+1)\cdot(6+1)\cdot(4+1)$

$= 8 \times 7 \times 5$

$= 280$.

$N = 24 = 2^3 \cdot 3^1$

1, 2, 3, 4, 6, 8, 12, 24 → Total = 8 div.

HCF	24
1	1×24
2	2×12
1	3×8
2	4×6

→ 4 ways.

T. Div. / 2

Sum of all divisors = $1 + 2 + 3 + 4 + 6 + 8 + 12 + 24$ ✓

$= (\overset{2^0}{1} + 2^1 + 2^2 + 2^3) \cdot (\overset{3^0}{1} + 3^1)$

$= 1 + 3 + 2 + 6 + 4 + 12 + 8 + 24$ ✓

$N = 2^7 \cdot 3^6 \cdot 5^4$

↳ No. of div. $= (7+1)\cdot(6+1)\cdot(4+1)$

$= 8 \times 7 \times 5$

$= 280$.

Sum of div. $= \left(2^0 + 2^1 + 2^2 + 2^3 + \ldots + 2^7\right)\cdot\left(3^0 + 3^1 + 3^2 + \ldots + 3^6\right)\cdot\left(5^0 + 5^1 + \ldots + 5^4\right)$

$N = 36 = 2^2 \cdot 3^2$

1, 2, 3, 4, 6, 9, 12, 18, 36 → 9 div.

36

1 × 36
2 × 18
3 × 12
4 × 9
6 × 6

5 ways.

$K = 2$

$2^{2-1} = 2^1$

$$\frac{\text{T. div.} + 1}{2}$$



DIVISOR PROBLEMS

$N = 2^{\alpha} \cdot 3^{\beta} \cdot 5^{\gamma} \cdot 7^{\delta} \ldots\ldots$

(I) Total number of divisors = $(\alpha + 1).(\beta + 1).(\gamma + 1).(\delta + 1)\ldots\ldots$

(II) Total number of proper divisors = $(\alpha + 1).(\beta + 1).(\gamma + 1).(\delta + 1)\ldots\ldots - 2$

(III) Sum of all the divisors = $(2^0+2^1+2^2+\ldots+2^{\alpha}).(3^0+3^1+\ldots+3^{\beta}).(5^0+5^1+\ldots+5^{\gamma})\ldots.$

(IV) Number of ways in which 'N' can be resolved as product of two factors

$= \left(\frac{\text{total divisors}}{2}\right)$, 'N' ≡ "is not a perfect square"

$= \left(\frac{\text{total divisors} + 1}{2}\right)$, 'N' ≡ "is a perfect square" (Hint: $\alpha, \beta, \gamma, \delta\ldots \rightarrow$ all are even)

(V) Number of ways in which 'N' can be resolved in product of two co-prime factors = 2^{k-1}, (where k = total no. of prime factors)

Bumper Question

If $N = 2^4.3^3.5^2.6^3$

1. (i) Total number of divisors
   (ii) total number of divisors which are perfect square
   (iii) Total number of divisors which are perfect cube.
2. (i) Total number of divisors divisible by 3.
   (ii) Total number of divisors divisible by 15
   (iii) Total number of divisors divisible by 12

Question

Find the number of arrangements of letters of the word EQUATION such that vowels occur in alphabetical order.

HW.

Question [JEE Mains-2026]

[Ans. : 126]

Two players A and B play a series of games of badminton, The player, who wins 5 games first, wins the series. Assuming that no game ends in a draw, the number of ways, in which player A wins the series is _____.

Question [JEE Mains-2026]

[Ans. : 660]

Let ABC be a triangle. Consider four points p_1, p_2, p_3, p_4 on the side AB, five points p_5, p_6, p_7, p_8, p_9 on the side BC, and four points $p_{10}, p_{11}, p_{12}, p_{13}$ on the side AC. None of these points is a vertex of the triangle ABC. Then the total number of pentagons, that can be formed by taking all the vertices from the points $p_1, p_2, \ldots, p_{13}$, is

Question

When repetition is allowed find the number of 7 digits numbers 0 to 9

(i) such that sum of digits is even

(ii) which are palindrome.

HW

DO IT BY YOURSELF (DIBY)

DIBY-38

There are 5 points P_1, P_2, P_3, P_4, P_5 on the side AB, excluding A and B, of a triangle ABC. Similarly there are 6 points $P_6, P_7, \ldots, P_{11}$ on the side BC and 7 points $P_{12}, P_{13}, \ldots, P_{18}$ on the side CA of the triangle. The number of triangles, that can be formed using the points $P_1, P_2, \ldots, P_{18}$ as vertices, is: [JEE Mains-2024]

(A) 751 (B) 776 (C) 796 (D) 771

[Ans. : (A)]

DIBY-39

A committee of 5 is to be chosen from a group of 9 people. Number of ways in which it can be formed if two particular persons either serve together or not at all and two other particular persons refuse to serve with each other, is

(A) 41 (B) 36 (C) 47 (D) 76

[Ans. : (A)]

DIBY-40

There are 15 players in a cricket team, out of which 6 are bowlers, 7 are batsmen and 2 are wicketkeepers. The number of ways, a team of 11 players be selected from them so as to include at least 4 bowlers, 5 batsmen, and 1 wicketkeeper, is

[Ans. : 777]

DIBY-41

Suppose that 20 pillars of the same height have been erected-along the boundary of a circular stadium. If the top of each pillar has been connected by beams with the top of all its nonadjacent pillars, then the total number of beams is:

(A) 190 (B) 170 (C) 180 (D) 210

[Ans. : (B)]

DIBY-42

Let $A = \{1, 2, 3, 4, 5, 6, 7\}$ and $B = \{3, 6, 7, 9\}$. Then the number of elements in the set $\{C \subseteq A : C \cap B \neq \phi\}$ is_____.

[JEE Mains-2022]

DIBY-43

If ${}^{2n}C_3 : {}^{n}C_3 = 10 : 1$, then the ratio $(n^2+3n):(n^2-3n+4)$ is [JEE Mains-2023]

(A) 65 : 37 (B) 2 : 1 (C) 36 : 16 (D) 27 : 11

[Ans. : (B)]

DIBY-44

If ${}^{n}P_r = {}^{n}P_{r+1}$ and ${}^{n}C_r = {}^{n}C_{r-1}$, then the value of r is equal to [JEEE Mains-2021]

(A) 1 (B) 4 (C) 2 (D) 3

[Ans. : (C)]

DIBY-45 or 37

A boy needs to select five courses from 12 available courses, out of which 5 courses are language course. If he can choose at most two language courses, then the number of ways he can choose five courses is [JEE Mains-2023]

[Ans. : 546]

DIBY-46

Consider a rectangle ABCD having 5, 7, 6, 9 points in the interior of the line segments AB, CD, BC, DA respectively. Let α be the number of triangles having these points from all 4 different sides as vertices and β be the number of quadrilaterals having these points from all 4 different sides as vertices. Then $(\beta - \alpha)$ is equal to [JEE Mains-2021]

(A) 795 (B) 1173 (C) 1890 (D) 717

[Ans. : (D)]

DIBY-47

There are 4 men and 5 women in Group A, and 5 men and 4 women in Group B. If 4 persons are selected from each group, then the number of ways of selecting 4 men and 4 women is _____. [JEE Mains-2024]

[Ans. : 5626]

DIBY-48

The letters of the word OUGHT are written in all possible ways and these words are arranged as in a dictionary, in a series. Then the serial number of the word TOUCH is [JEE Mains-2023]

(A) 89 (B) 84 (C) 86 (D) 79

[Ans. : (A)]

DIBY-49

All words, with or without meaning, are made using all the letters of the word MONDAY. These words are written as in a dictionary with serial numbers. The serial number of the word MONDAY is [JEE Mains-2023]

(A) 327 (B) 328 (C) 324 (D) 326

[Ans. : (A)]

DIBY-50

All the letters of the word PUBLIC are written in all possible orders and these words are written as in a dictionary with serial numbers. Then the serial number of the word PUBLIC is [JEE Mains-2023]

(A) 576 (B) 578 (C) 580 (D) 582

[Ans. : (D)]

DIBY-51

The letters of the word 'MANKIND' are written in all possible orders and arranged in serial order as in an English dictionary. Then the serial number of the word 'MANKIND' is [JEE Mains-2022]

[Ans. : 1492]

DIBY-52

All the letters of the word "GTWENTY" are written in all possible ways with or without meaning and these words are written as in a dictionary. The serial number of the word "GTWENTY" is [JEE Mains-2024]

[Ans. : 553]

DIBY-53

60 words can be made using all the letters of the word BHBJO, with or without meaning. If these words are written as in a dictionary, then the 50th word is:

[JEE Mains-2024]

(A) JBBOH (B) OBBJH (C) OBBHJ (D) HBBJO

[Ans. : (B)]

DIBY-54

If all the words with or without meaning made using all the letters of the word "NAGPUR" are arranged as in a dictionary, then the word at 315th position in this

[JEE Mains-2024]

(A) NRAGUP (B) NRAPUG (C) NRAPGU (D) NRAGPU

[Ans. : (C)]



Homework

Re-attempt all Class Examples

DPP & DIBY

It is never about end result

It's always about Journey

Thank You

#futureIITians